

MASINASIN AGDM, AB, Canada

WMO: 713450

Lat: 49.1367N

Lon: 111.6517W

Elev: 948

StdP: 90.44

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.1	-23.8	-30.5	0.2	-27.0	-27.1	0.4	-23.7	15.5	3.7	13.6	1.9	3.3	220	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	33.0	16.1	31.1	15.6	29.0	15.3	18.2	27.4	17.2	27.3	16.3	26.3	4.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.1	12.0	21.0	13.7	11.0	19.9	12.6	10.2	18.7	55.2	27.4	51.8	27.4	49.0	26.6	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	11.5	10.1	-32.3	36.7	2.4	1.7	-34.0	37.9	-35.4	38.9	-36.7	39.8	-38.4	41.1		
(5)			-32.5	19.9	2.3	1.4	-34.1	20.9	-35.5	21.7	-36.8	22.5	-38.5	23.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-4.8	-5.8	0.3	5.8	11.3	15.4	19.8	18.6	13.6	6.7	0.2	-5.7		
	DBStd	10.92	8.35	8.59	7.25	5.04	4.44	3.45	3.29	3.78	4.96	5.30	7.47	8.13		
	HDD10.0	2314	459	441	302	141	38	2	0	1	21	126	297	487		
	HDD18.3	4518	718	674	558	377	221	99	22	43	155	361	544	745		
	CDD10.0	984	0	0	2	13	78	166	303	267	129	24	3	0		
	CDD18.3	147	0	0	0	0	3	12	67	52	13	1	0	0		
	CDH23.3	2621	0	0	0	9	98	246	1097	872	288	11	0	0		
	CDH26.7	1017	0	0	0	1	19	63	466	364	102	1	0	0		
(14) Wind	WSAvg	4.8	5.5	4.7	5.1	5.1	4.8	4.6	4.0	3.9	4.4	5.0	5.5	5.4		
(15) Precipitation	PrecAvg	358	14	12	18	33	47	81	37	36	34	23	15	12		
	PrecMax	568	41	33	43	70	98	169	135	109	77	80	30	30		
	PrecMin	213	2	1	0	4	8	13	3	3	0	4	0	0		
	PrecStd	79	8	9	13	17	24	38	29	25	21	18	8	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	13.8	18.4	24.6	29.0	32.2	35.8	34.9	32.4	25.0	18.5	11.7		
		MCWB	5.7	5.4	8.4	11.5	14.0	17.0	17.2	16.1	14.9	12.6	8.7	5.0		
	2%	DB	8.6	9.7	15.3	20.5	26.1	28.6	33.6	32.7	29.8	20.8	14.5	8.5		
		MCWB	3.9	4.0	6.9	9.7	13.1	15.8	16.8	15.5	14.0	10.5	7.5	3.9		
	5%	DB	6.7	7.1	12.8	17.7	23.5	26.2	31.5	30.8	27.0	17.7	11.9	6.4		
		MCWB	2.8	2.4	5.7	8.1	11.9	15.0	16.5	14.9	13.2	9.3	5.9	2.5		
	10%	DB	5.1	5.0	10.3	15.0	20.9	24.1	29.3	28.5	23.6	15.1	9.4	4.6		
		MCWB	1.8	1.5	4.5	6.8	10.9	13.9	16.0	14.7	12.2	8.0	4.5	1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	6.3	9.1	11.7	14.6	18.8	20.1	18.1	16.4	13.5	10.1	6.2		
		MCDB	10.3	12.3	17.3	22.7	26.5	26.9	30.5	26.8	26.7	22.8	17.2	10.6		
	2%	WB	4.3	4.2	7.5	10.1	13.4	17.2	18.8	17.0	14.9	11.3	7.7	4.2		
		MCDB	8.2	9.3	14.2	19.8	24.1	26.5	27.5	28.5	26.4	19.3	13.9	8.1		
	5%	WB	3.0	2.8	6.1	8.5	12.5	15.8	17.7	16.1	13.8	9.7	6.2	2.7		
		MCDB	6.2	6.8	12.3	16.6	21.8	24.2	27.4	27.3	25.2	16.4	11.3	6.0		
	10%	WB	1.9	1.6	4.6	7.1	11.5	14.7	16.9	15.3	12.7	8.4	4.7	1.5		
		MCDB	4.9	4.8	9.9	14.0	19.5	22.5	26.9	26.0	23.3	14.4	9.1	4.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	11.1	11.8	12.8	13.2	13.2	16.2	15.8	14.3	12.5	11.0	9.9		
		MCDBR	10.3	13.2	15.4	17.9	17.9	17.3	19.9	20.3	20.1	16.3	13.2	10.5		
		MCWBR	6.8	8.4	8.6	8.8	7.6	7.2	7.0	7.2	8.1	8.2	7.5	7.2		
		MCDBR	9.9	12.4	14.6	17.0	16.0	15.7	16.5	17.2	18.2	15.4	12.9	10.3		
	5% WB	MCWBR	7.0	8.3	8.5	8.6	7.1	7.2	6.8	6.6	7.6	8.0	7.8	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.263	0.282	0.327	0.343	0.340	0.342	0.362	0.322	0.285	0.258	0.247		
	taud		2.463	2.494	2.499	2.371	2.370	2.431	2.436	2.351	2.435	2.527	2.510	2.446		
	Ebn at Noon		832	911	943	921	916	919	911	874	881	868	823	792		
	Edh at Noon		60	74	87	111	116	110	108	112	93	71	56	53		
(44) All-Sky Solar Radiation	RadAvg		1.28	2.25	3.53	4.90	5.83	6.51	7.11	5.72	4.08	2.55	1.45	1.00		
	RadStd		0.11	0.19	0.28	0.31	0.45	0.40	0.37	0.36	0.32	0.24	0.10	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page